

A DATA SCIENCE APPROACH TO MEDICAL IMAGING AND HEALTHCARE DATA ANALYSIS

Naina Morya^{*1}, Rohit Kumar^{*2}

^{*1}Computer Science Student At St. Soldier Group Of Institution, Jalandhar, Punjab, India.

^{*2}Software Engineer At Rising Stars, Jalandhar, Punjab, India.

DOI: <https://www.doi.org/10.56726/IRJMETS63096>

ABSTRACT

This research identifies the data science tools or techniques to assist healthcare with the healthcare data. Our objective is to find the optimal tools to analyze the data effectively. In this paper, we have found suitable data science tools to be used with CT scans, MRI, and CAD to conduct analysis. Our results show that OpenCv is the common tool used whenever data is in image format. However, the data training model is required to identify the disorder type correctly. Image data can be further converted into table format with the help of data integration. To deal with the data in table Pyspark, clustering, and AI methods are used to complete the analysis. To finalise the report PowerBI is used to make the data into graphical form.

Keywords: Data Science; Healthcare; Data Integration; Image Analysis; Predictive Analysis; CT Scan; MRI; Power BI; Clustering.

I. INTRODUCTION

Data Science is an interdisciplinary field combining data, computing, and machine learning techniques to examine and explain huge quantities of data [1]. It has emerged as a crucial tool in different businesses, especially in healthcare, where large amounts of clinical data are created daily, requiring modern data handling techniques[2]. The ability of Data Science to transform healthcare sources from its capacity to filter significant observations from mixed and complicated data collections; including medical visualization, EHRs (Electronic Health Records), and genetic data[3].

In healthcare services, Data Science enhances treatment-oriented problem-solving, boosts diagnostic reliability, and forecasts patient results as an illustration predictive analysis allows healthcare specialists to predict disorder development and assign assets more effectively [4]. Furthermore, machine learning methods are implemented in imaging diagnostics to discover abnormalities such as cancerous development, thus assisting in early detection [5] Therefore, data science is necessary to handle both functional and medical obstacles in healthcare[6].

Regardless of its capability, combining Data Science with medical care suffers few complications. One of the remarkable challenges is the many tools and techniques used in healthcare analytics, which are dominant in a divided framework where information is regularly isolated and conflicting[7]. These dividing, challenges the normalization of data structures and integration methods, restricting the utilization of Data Science [8]. In addition, worries about data protection and safety must be precisely directed due to the susceptible existence of healthcare datasets[9].

Data Science deals with different cures to the complications of division and data integration in clinical care. By utilizing modern data integration strategies, Data Science can merge different data collections, supporting vast analysis and enhancing compatibility over healthcare frameworks[10]. As an illustration tools such as PowerBI help in creating perceptive documents based on complicated healthcare data while frameworks like PySpark enable managing huge data collections[11]. Furthermore, machine learning and predictive analysis techniques can assist address inequalities in analytical reliability and enhance unique care techniques [12]. This paper will emphasize these resolutions by concentrating on precise healthcare applications, like the utilization of Data Science tools in oncology and cardiology.

The scope of this paper is restricted to examining the applications of Data Science in chosen fields of healthcare like imaging (MRI, CT scan), cardiology, and predictive analysis in disorder management. Meanwhile, the medical sector is vast and mixed, this paper focuses on detecting and analyzing the most suitable tools and

techniques for these exact applications, instead of delivering a comprehensive review of the whole medical field[3]. Upcoming efforts could extend these discoveries to contain other medical expertise.

II. LITERATURE REVIEW

Data Science has transformed into a crucial element of current healthcare because of its capacity to analyze complicated data collection and create applicable observations. This segment delivers a summary of the wide tools and strategies of data science, additionally, their operations in healthcare, illustrating in which these tools are executed to figure out healthcare obstacles.

2.1 General Tools and Techniques of Data Science

Data Science appoints an extensive scope of tools and techniques to analyze complicated data obstacles, as defined in our research. Python which is broadly utilized for data manipulation and numerical analysis. Python is especially trendy due to its adaptable libraries, as an illustration, pandas are used for data manipulation and NumPy is used for numerical calculations. Pandas are particularly beneficial for supervising planned data in templates like Excel, CSV, and SQL databases, permitting, healthcare scientists to organize and operate data productively. Simultaneously, NumPy assists computational data, that are crucial for numerical analysis in healthcare[13].

Another crucial part of Data Science is the use of huge data structures like Apache Spark and Apache Hadoop, which are created to manage big data by allocating the algorithmic task across several machines[14]. These structures are crucial in fields with enormous data, containing healthcare. Furthermore, machine learning methods such as Neural Networks, Random Forests, and Support Vector Machines (SVM) are extensively used for sorting, regression, and predictive analytics[15].

2.2 Data Science Tools Used in Healthcare

Data science tools play a vital role in converting healthcare by enhancing analysis care, and patient supervision. In medical visualization, tools such as OpenCV and TensorFlow are utilized for Computer-Aided Systems (CAD). OpenCV is broadly implemented for image analysis functions, like feature extraction and partitioning, that assist radiologic specialists in recognizing irregularities in radiologic images such as MRIs and CT scans[16]. Tensor Flow, However, enables deep learning functions by managing massive image data structures, containing DICOM, necessary for precise analytical testing[17].

Clustering strategies comprised of K- Means, perform an essential role in disorder discovery and patient fragmentation. K-means clustering teams up patients on behalf of collective attributes, like threat discovery for cardiovascular disease or cancer, hence assisting customized treatment and prior ailment discovery. In cardiology for detecting patients at threat of heart disease, facilitating personalized cures is done using clustering techniques [18].

Data integration tools such as PySpark effectively process a huge quantity of data of patient details from electronic health records (EHRs) and merge with genetic data. This integration offers a comprehensive perspective of patient health, improving problem-solving in complicated disorder management[14]. PySpark efficiently processes huge amounts of patient information, assisting in complete analysis and enhancing coherence over healthcare frameworks[15].

Predictive analytics is another remarkable proposal of data science in healthcare. Tools like Power BI are employed to illustrate predictive analysis outcomes, permitting healthcare services to anticipate capable health events such as disorder development or cure results. In cardiology, predictive analysis assists in predicting cardiovascular disorders, directing doctors to develop defensive treatment methods [19]. Likewise, in pulmonology, predictive patterns help in observing lung task decrease in chronic situations, offering prior alerts to healthcare doctors[17].

Sentiment analysis, like TextBlob, is employed to investigate patient reviews, thus enhancing healthcare resources. TextBlob operates printed data from patient feedback, classifying it through positive, negative, and neutral emotions. This investigation assists in healthcare companies recognizing patient enduring and creating essential enhancement attention standards [20].

Bioinformatics tools, containing PySpark and Pandas, are typically utilized for combining multi-omics data structures such as BAM and FASTQ, assisting in genetic investigations and customized treatment. These tools

help in the analysis of complicated organic data, managing new features into disorder frameworks, and aptitude healing calming goals[21].

Overall, the deployment of these data science tools or techniques in healthcare has significantly improved analytical precision, cure personalization, and patient attention. In contrast, obstacles such as data security, treatment concerns, and the requirement for uniform data structures persist frequently and necessitate current focus to increase the advantages of data science in healthcare[14][15].

III. TOOLS AND TECHNIQUES OF DATA SCIENCE IN HEALTHCARE APPLICATION

3.1 Image Analysis

3.1.1 Purpose:

Image analysis in healthcare aims to help in detecting situations by examining medical images such as MRI and CT scans. It assists in identifying crucial challenges e.g. tumors, bone injuries, and other internal anomalies, assisting in timely interference[5].

- **CAD (Computer-Aided System):** CAD (Computer-Aided System): Computer-Aided Systems merge image analysis, pattern analysis, and smart technology. It mainly concentrates on categorizing harmless and cancerous growth, depending on patterns gained out of previously verified instances.
- **CT (Computed Tomography) Scan:** CT scan has gradually been used in medical facilities, providing image processing problems in the cross-axis plane as illustrated below:

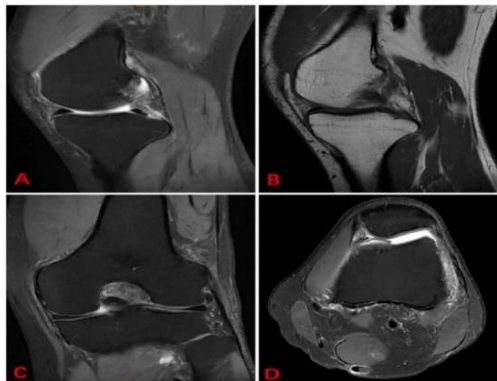


Figure 1: Various CT Scan pictures of distressing knee joint wound.

- **MRI (Magnetic Resonance Imaging):** MRI successfully identifies complicated breaks and allows imaging over multiple angles as described below.



Figure 2: Various MRI pictures of distressing knee joint wounds

3.1.2 Importance:

MRI and CT scans are required for recognizing internal hurdles. MRI offers definitive images of smooth tissues, while CT scan offers precise cross-sectional views of organs and bones. These strategies are crucial for precise analysis and treatment organization [22].

3.1.3 Tools: Open CV

For medical image computing, Open CV is used. It implements techniques such as edge recognition and segmentation to CT and MRI visions, optimizing their distinctness and emphasizing abnormal areas, assisting radiologists in recognizing them[11]. Open CV operates image structures like PNG, JPEG, BMP, and DICOM that

are mainly utilized in medical utilization[23]. Data present in images is further transformed into the tabular layout for enhanced analysis utilizing the integration method, which improves or increases data availability clarity over various critical tools[16].

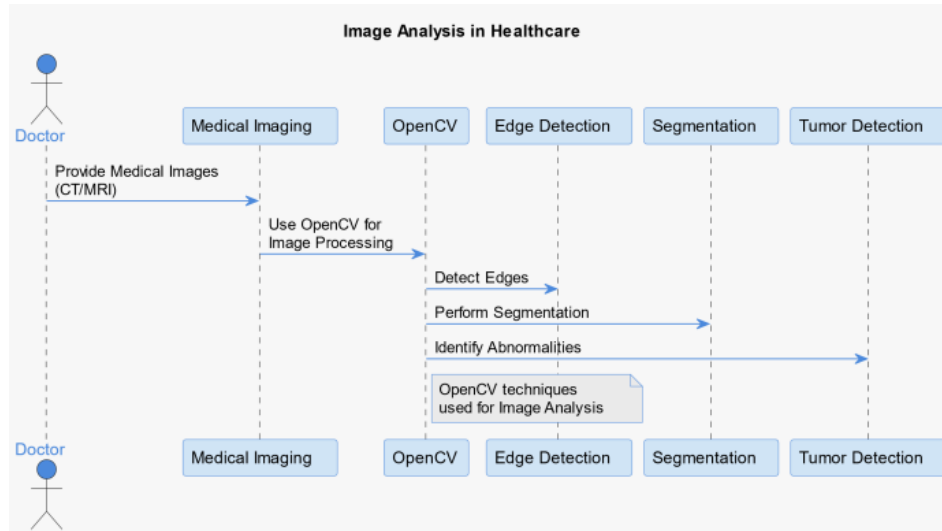


Figure 3: Workflow of Image Analysis in Healthcare Using OpenCV

3.2 Clustering for Disease Detection

3.2.1 Purpose:

Clustering assists in combining patients on behalf of their characteristics, like threat factors for cardiovascular disease or cancer. It facilitates healthcare physicians to categorize patients for enhanced recognition and treatment.

- **Cardiology:** K-process clustering algorithm employed to detect heart disorder patients.
- **Oncology:** The Clustering algorithm is also used to detect colon tumors[24]

3.2.2 Importance:

By classifying patients into definite risk collections, the clustering algorithm assists in premature identification and customized treatment, which enhances patient results and resource control[25].

3.2.3 Tool: K-means

K-means clustering group patients' records into clusters, assisting in finding threatening patients with disorders like cardiovascular diseases. This method utilizes patterns in the data sets to categorize patients on behalf of their clinical records and health records[24]. It uses numerical and enumerated data in Excel, CSV, and database layouts for patient fragmentation and threat analysis [26].

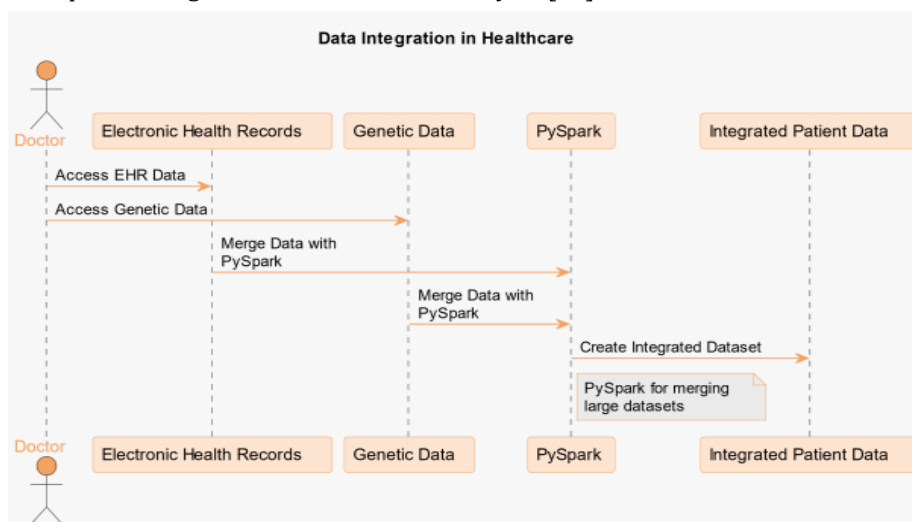


Figure 4: Clustering Patients for Disease Detection Using K-Means

3.3 Data Integration:

3.3.1 Purpose:

Data integration aims to merge data from different resources like EHRs, hereditary data, and health records. This gives a complete sight of the patient's health for enhanced judgment.

- **Bioinformatics:** Bioinformatics is a multidisciplinary field of research that link to analytics, biochemistry, arithmetic, and computing to figure out health issues as described below [11].

3.3.2 Importance:

Integrating huge and heterogeneous healthcare data collections allows for comprehensive patient treatment and customized prescriptions, enhancing treatment results and healthcare productivity[27].

3.3.3 Tools: PySpark

PySpark operates and merges enormous data collections in healthcare, and permits productive diagnosis of vast health data, from electronic health records to genetic data. It assists in current data handling and patient records integration throughout frameworks[11]. It was developed to operate extensive data collections in parquet, JSON, CSV, ORC, Avro, and Hive tables[15][28].

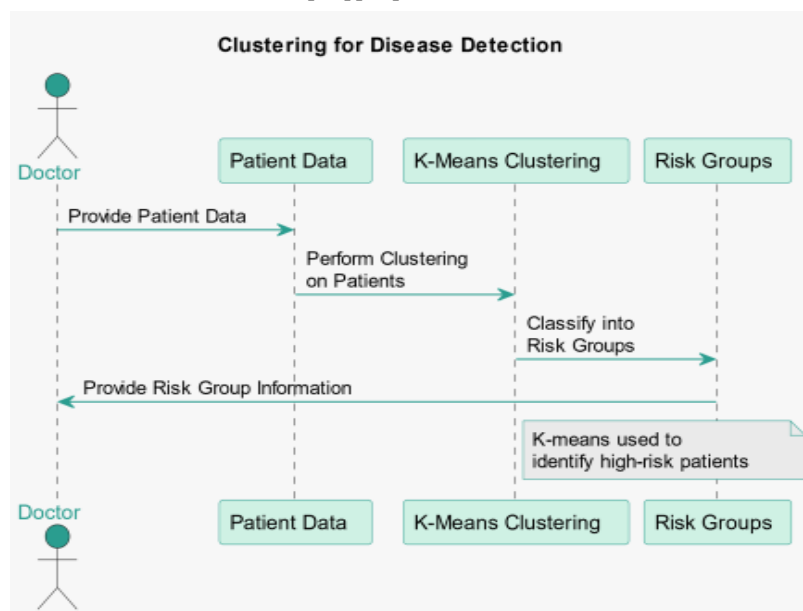


Figure 5: Data Integration Process in Healthcare Using PySpark

3.4 Predictive Analytics

3.4.1 Purpose:

Predictive analytics aims to predict future fitness events, like disorder development or treatment results, on behalf of modern and past patient data.

- **Cardiology:** Predictive analysis is utilized in cardiology to forecast the threat of heart diseases, and notify cure conclusions.
- **Pulmonology:** In pulmonology, observation and forecasting lung behavior deterioration, especially in illnesses like chronic lung disease.
- **Hematology:** Predictive analysis is utilized in Hematology to analyze and regulate blood diseases like anemia and clotting problems.
- **Sleep Disorders:** predictive analysis assists in recognizing sleep problems and their effect on physical health as illustrated below[29].

3.4.2 Importance:

This assists healthcare physicians make knowledgeable choices, improving treatment strategies, and stopping harmful health results, thus enhancing entire patient concerns[4].

3.4.3 Tools: Power BI

Power BI assists in visualizing predictive analytics outcomes, showing patterns and threat factors in simple-to-

recognize dashboards. This permits physicians to fast examine patients' threat levels and take protective actions[29]. It employs an extensive scope of data entries containing Excel, JSON, SQL, CSV, and APIs for optical documentation[19].

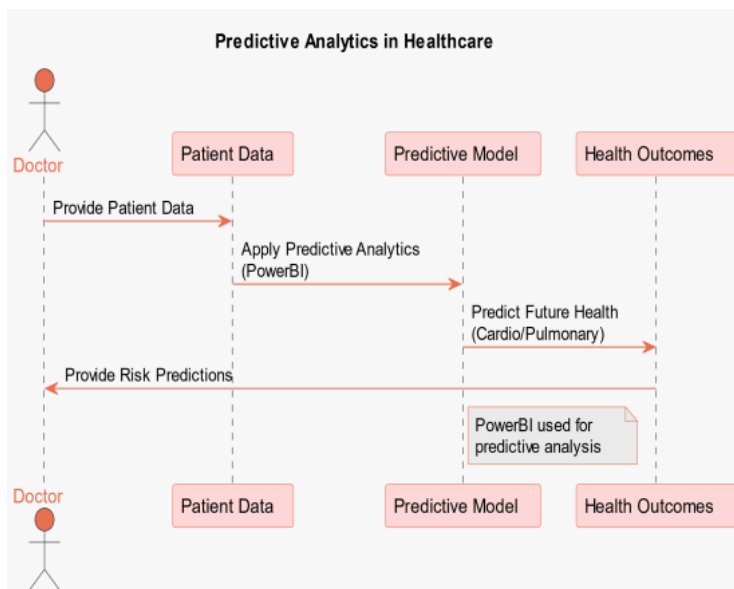


Figure 6: Predictive Analytics in Healthcare Using PowerBI

3.5 Sentiment analysis in patient feedback

3.5.1 Purpose: sentiment analysis aims to examine patients' reviews to evaluate contentment and find areas for enhancing healthcare expertise.

- **Sentiment Analysis (Various Fields of Medical):** Sentiment analysis utilizes ML (Machine Learning) on client feedback. The outcome of SA (Sentiment Analysis) is employed to determine the “evaluation score” for health management facilities.

3.5.2 Importance: By recognizing patients' emotions, healthcare doctors can improve the efficiency of concern, enhance interaction, and raise patients' happiness.

3.5.3 Tools: TextBlob

Text Blob is utilized for sentiment analysis by handling patient feedback and classifying it into positive, negative, or neutral gratitudes. This aids healthcare institutions in understanding patient care and enhances their assistance[12]. It operates with unsystematic text data in structures like CSV, Plain text, and JSON for sentiment analysis[20].

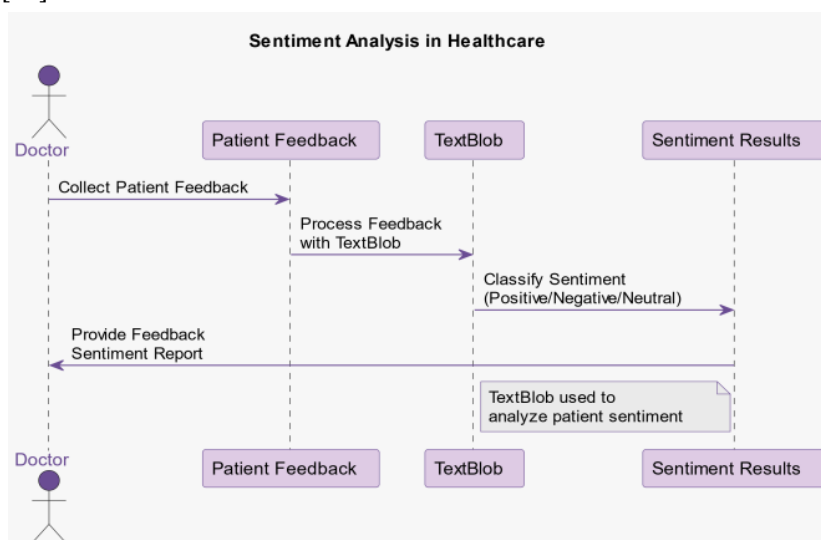


Figure 7: Sentiment Analysis of Patient Feedback Using TextBlob

IV. RESULT

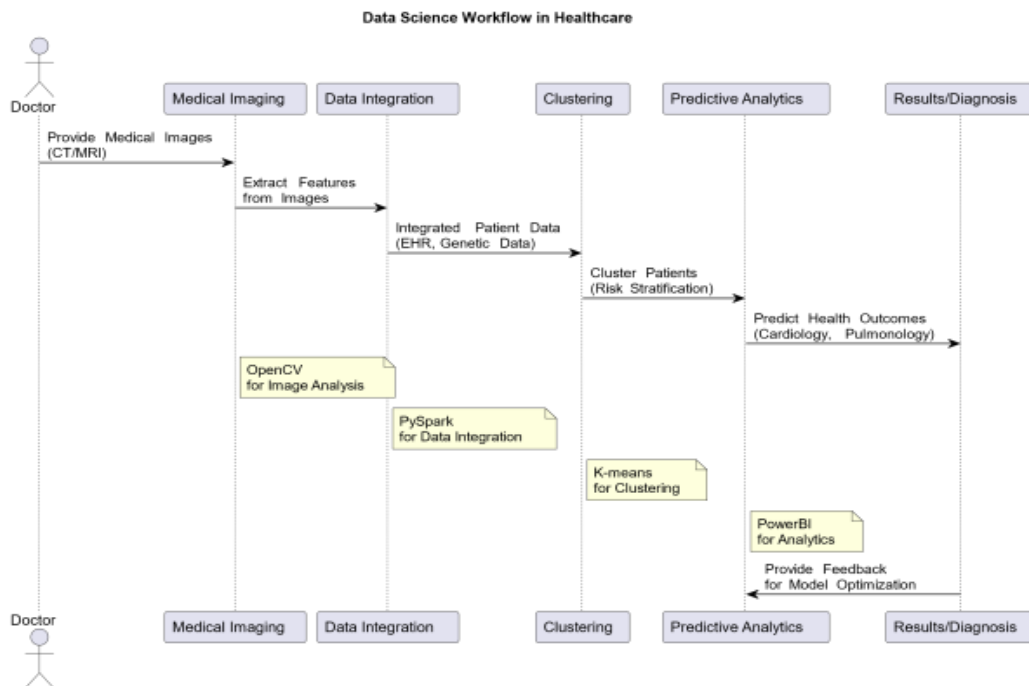


Figure 8: Overview of Data Science Techniques in Healthcare

Table 1: Highlights of data science techniques and their application in different sectors of the Medical

Application	Tools/Techniques	Benefits
CAD	Image Analysis (Uses Open CV)	Medical Device Design Surgical Planning,3D printing
CT Scan		Diagnostic Imaging, Visualization
MRI		High Resolution images, Advanced Functional imaging capabilities
Oncology	Clustering (Uses K-means)	Personalized treatment, uncover hidden patterns, Enhance cancer research
Cardiology		Improve diagnosis, Risk Stratification
Bioinformatics	Data Integration (Uses Pyspark)	Comprehensive understanding, insights into complex diseases, Drug discovery
Cardiology	Predictive Analytics (Uses Power BI)	Optimizing treatment plans, Identifying high-risk patients, and Early detection of diseases
Pulmonology		Early detection of respiratory failure, Predicting asthma
Hematology		Predicting bleeding risk, blood disorders
Sleep Disorder		Predicting sleep apnea, Sleep-

		related disorders
Sentiment Analysis (Various field of Medical)	Machine Learning (Text Blob)	Clinical trial, Patient engagement, Medical literature analysis

V. CONCLUSION

The different kinds of data science tools like open CV (computer vision), Data Integration uses pyspark, predictive Analysis uses BI, and many more are employed broadly in various fields of medicine like cardiology, pulmonology, etc. Table 1 shows the list of data science techniques or tools which is used in several sectors of medical or health care. Data science tools help in analysing disorders in the human body like injuries, blood disorders, lung disease, sleep problems, and many more. Thus, it helps the doctors to identify patient's disease after proper analysis with test reports integrated with AI. These tools are also used in different machines such as health management areas like CT scans and MRI for analysing the image of patient injuries and disorders. AI techniques like machine learning, data integration, and clustering provide us with proper diagnosing of several diseases and integrating patient data from various sources as well as uncover hidden patterns in diseases. In the future, these different tools will be more effective and faster. Even if data science tools and AI tools make predictions based on its analyses, there may be some malfunctioning of instrument therefore we have to use multiple AI methods to ensure that we are making the right prediction as our prediction is directly related to the user's health. These data tools can identify all kinds of disorders, to improve accuracy, we can also target some of the possible diseases. In the future, there will be large data training sets that help AI to analyse diseases in less time. To analyse medical problems, we need advanced GPUs to analyse the data. However, by using algorithms we can reduce the GPU requirement.

VI. REFERENCES

- [1] F. Provost and T. Fawcett, "Data Science and its Relationship to Big Data and Data-Driven Decision Making," Big Data, vol. 1, no. 1, pp. 51–59, 2013.
- [2] V. Dhar, "Data science and prediction," Commun. ACM, vol. 56, no. 12, pp. 64–73, 2013.
- [3] B. Shickel, P. J. Tighe, A. Bihorac, and P. Rashidi, "Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record (EHR) Analysis.," IEEE J. Biomed. Heal. informatics, vol. 22, no. 5, pp. 1589–1604, Sep. 2018.
- [4] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley, "Deep learning for healthcare: review, opportunities and challenges.," Brief. Bioinform., vol. 19, no. 6, pp. 1236–1246, Nov. 2018.
- [5] A. Esteva et al., "A guide to deep learning in healthcare," Nat. Med., vol. 25, no. 1, pp. 24–29, 2019.
- [6] F. Jiang et al., "Artificial intelligence in healthcare: Past, present and future," Stroke Vasc. Neurol., vol. 2, no. 4, pp. 230–243, 2017.
- [7] W. Raghupathi and V. Raghupathi, "Big data analytics in healthcare: promise and potential.," Heal. Inf. Sci. Syst., vol. 2, p. 3, 2014.
- [8] H. M. Krumholz, "Big data and new knowledge in medicine: the thinking, training, and tools needed for a learning health system.," Health Aff. (Millwood), vol. 33, no. 7, pp. 1163–1170, Jul. 2014.
- [9] B. Mittelstadt, "Ethics of the health-related internet of things: a narrative review," Ethics Inf. Technol., vol. 19, no. 3, pp. 157–175, 2017.
- [10] A. Torab-Miandoab, T. Samad-Soltani, A. Jodati, F. Akbarzadeh, and P. Rezaei-Hachesu, "A unified component-based data-driven framework to support interoperability in the healthcare systems," Heliyon, vol. 10, no. 15, p. e35036, 2024.
- [11] N. Iqbal and P. Kumar, "From Data Science to Bioscience: Emerging era of bioinformatics applications, tools and challenges," Procedia Comput. Sci., vol. 218, no. 2022, pp. 1516–1528, 2022.
- [12] P. M. Madan, M. R. Madan, and D. P. Thakur, "Analysing The Patient Sentiments in Healthcare Domain Using Machine Learning," Procedia Comput. Sci., vol. 238, pp. 683–690, 2024.
- [13] K. Wang, L.; Yu, H.; Xu, "Data manipulation and analysis with Pandas in healthcare research," Data Sci. Med., vol. 4, no. 2, pp. 98–108, 2020.
- [14] P. T. Nguyen, T. H.; Tran, L. Q.; Le, "Big data processing in healthcare using Apache Spark: Formats and techniques," Int. J. Big Data, vol. 7, no. 4, pp. 201–209, 2019.

- [15] X. Zhang, R.; Wang, F.; Lin, "PySpark in healthcare: Efficient data processing for large-scale analytics," J. Healthc. Big Data, vol. 8, no. 3, pp. 233–242, 2019.
- [16] M. Li, Y.; Chen, X.; Zhao, "Processing DICOM images using OpenCV: A practical approach in healthcare diagnostics," J. Imaging Sci., vol. 12, no. 6, pp. 1024–1035, 2018.
- [17] T. Johnson, S. P.; Patel, M.; Gomez, "TensorFlow in medical image analysis: Handling DICOM data for diagnostics," IEEE Trans. Med. Imaging, vol. 41, no. 7, pp. 1567–1574, 2022.
- [18] H. Lee, C.; Park, J.; Kim, "Application of K-means clustering in disease risk prediction using clinical data," Healthc. Data Sci. J., vol. 9, no. 2, pp. 120–132, 2020.
- [19] P. Brown, A. M.; Thomas, J.; Roberts, "Data integration using Power BI in healthcare: Enhancing visual analytics and decision-making," J. Med. Informatics, vol. 28, no. 4, pp. 178–185, 2021.
- [20] R. Smith, J.; Davis, L.; Adams, "Sentiment analysis of patient feedback using TextBlob: An approach for healthcare improvement," J. Healthc. Anal., vol. 15, no. 1, pp. 55–64, 2021.
- [21] V. Davis, R. J.; Iqbal, Z.; Kumar, "Integration of multi-omics data in bioinformatics: Applications in healthcare," Bioinforma. Comput. Biol., vol. 35, no. 3, pp. 295–307, 2020.
- [22] M. Kong and F. Wang, "Comparative analysis of computer-aided imaging collaboration: MRI versus CT for detection of knee joint injuries in athletes," J. Radiat. Res. Appl. Sci., vol. 17, no. 3, p. 100960, 2024.
- [23] M. Li, Y. Jiang, Y. Zhang, and H. Zhu, "Medical image analysis using deep learning algorithms," Front. Public Heal., vol. 11, no. November, pp. 1–28, 2023.
- [24] J. Santos-Pereira, L. Gruenwald, and J. Bernardino, "Top data mining tools for the healthcare industry," J. King Saud Univ. - Comput. Inf. Sci., vol. 34, no. 8, Part A, pp. 4968–4982, 2022.
- [25] N. Kumar, P. and Iqbal, "Machine learning approaches for precision medicine," IEEE Access, vol. 8, no. 160960–160974, 2020.
- [26] J. Lu, "Data science in the business environment: Insight management for an Executive MBA," Int. J. Manag. Educ., vol. 20, no. 1, Mar. 2022.
- [27] D. Mietchen, "Data sharing in public health emergencies," Int. J. Infect. Dis., vol. 53, pp. 35–36, 2016.
- [28] H. Jahani, R. Jain, and D. Ivanov, "Data science and big data analytics: a systematic review of methodologies used in the supply chain and logistics research," Ann. Oper. Res., 2023.
- [29] T. A. Pearson et al., "Precision Health Analytics With Predictive Analytics and Implementation Research: JACC State-of-the-Art Review," J. Am. Coll. Cardiol., vol. 76, no. 3, pp. 306–320, 2020.